

Reading a research claim

Project: Reproducible AI Research Study

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

A question that evidence can answer

Without opening the old lesson, explain one key idea from A question that evidence can answer. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Reading a research claim. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Reading a research claim. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Report claims universal speed after median2 seconds versus3 on four local fixtures. Audit scope.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision

rule and output.

My reasoning and evidence

6. W2 Trace the worked case

A fictional report says method B is better after 6/8 supported answers versus baseline 5/8 on the same constructed cases. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: A numerical improvement can be real in the sample while the broader claim remains unsupported. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

Report claims universal speed after median2 seconds versus3 on four local fixtures. Audit scope.

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled reading a research claim evidence sheet to Reproducible AI Research Study. Show how you can explain the mechanism and terminology of reading a research claim. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Reading a research claim

Read a claim with its denominator. A fictional paper headline says a prompt is twice as effective. Its table reports four correct cases instead of two.

Baseline: 2/10 correct. New prompt: 4/10 on the same ten cases. Scoring is fixed.

AI ACADEMY / AI-7/2

Correctness on ten shared cases

correct answers out of 10

Baseline



New prompt



Bar length shows the supplied value; the marked baseline is zero.

Reading a research claim

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Find the measured outcome

The table counts correct answers, not learning or satisfaction.



2. Calculate both changes

20% → 40% is twice the rate and a 20-percentage-point increase.



3. Check claim scope

These ten cases support a descriptive comparison, not universal doubling.

CHANGE ONE THING

The headline changes to "every learner improves twice as much".

Explain the changed result using the matching lesson.

Calculate the baseline and new rates for 2/10 and 4/10. Explain both twice the rate and the percentage-point change, and say why neither measures every learner's improvement.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

13. Project portfolio milestone

Project: Reproducible AI Research Study

Current milestone: Frame the need. Name the user, the problem, allowed inputs and a safe simple starting method.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Read a claim with its denominator. A fictional paper headline says a prompt is twice as effective. Its table reports four correct cases instead of two.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? These ten cases support a descriptive comparison, not universal doubling.

1. Check claim scope
2. Calculate both changes
3. Find the measured outcome

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Check claim scope These ten cases support a descriptive comparison, not universal doubling. A research claim compresses a method and result. Unpack the measured variable, denominator and comparison before accepting the headline.

Practice 2: explain the deciding evidence

Changed-case practice: The headline changes to "every learner improves twice as much". Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.
3. That learner claim is unsupported by these answer counts.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: That learner claim is unsupported by these answer counts. The outcome and experimental unit have changed in the wording without new evidence.

Repair a missing building block

Review A question that evidence can answer (ai-7/1). Goal: Explain the mechanism and terminology of a question that evidence can answer. Short practice: Which proposed diagram question has a measurable outcome? Name its versions, six-case denominator and fixed scoring, then explain why giving only B easier tasks spoils the comparison.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
